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Perceptron XOR Problem

A digital alarm has two binary sensors: x_1 and x_2

The alarm sound turn on when exactly one sensor is ON. This is called XOR operation

The Perceptron uses the following formula.

$$z = w_1x_1 + w_2x_2 + b$$

The active function is.

$$H(z) = \begin{cases} 1, & z \geq 0 \\ 0, & z < 0 \end{cases}$$

Here

x_1, x_2	Input Sensors
w_1, w_2	Weights of Inputs
b	Bias
z	Weighted sum plus bias
$H(z)$	Final output

(1) Write all four Inequalities.
The truth table is:

x_1	x_2
0	0
0	1
1	0
1	1

Rule:- If the output should be 0 then $z < 0$. if the output should be 1, then $z \geq 0$

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Case-1. $x_1 = 0, x_2 = 0, y = 0$

$$z = w_1(0) + w_2(0) + b = b$$

Therefore

$$b < 0$$

Case 2.

$$x_1 = 0, x_2 = 1, y = 1$$

$$z = w_1(0) + w_2(1) + b = w_2 + b$$

Therefore

$$w_2 + b \geq 0$$

Case 3.

$$x_1 = 1, x_2 = 0, y = 1$$

$$z = w_1(1) + w_2(0) + b = w_1 + b$$

Therefore

$$w_1 + b \geq 0$$

Case 4. $x_1 = 1, x_2 = 1, y = 0$

$$z = w_1 + w_2 + b$$

Therefore

$$w_1 + w_2 + b < 0$$

These are the four required inequalities.

(2) Prove that a single perceptron can not classify XOR correctly

We need to prove that no values of w_1, w_2 and b can satisfy all four inequalities at the same time.

Step 1: Use the second inequality

$$w_2 + b \geq 0$$

Therefore

$$w_2 \geq -b$$

Step 2: Use the inequality

$$w_1 + b \geq 0$$

Therefore

$$w_1 \geq -b$$

Step 3: Add both inequality

$$w_1 + w_2 \geq -2b$$

Adding b to both sides gives

$$w_1 + w_2 + b \geq -b$$

Step 4: Use the first inequality.

We know that

$$b < 0$$

Therefore

$$-b > 0$$

So

$$w_1 + w_2 + b > 0$$

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Step 5 find the contradiction.
But the fourth inequality requires

$$w_1 + w_2 + b < 0$$

The same expression can not be both greater than zero and less than zero

Therefore the four inequalities can not be satisfied simultaneously

Conclusion

A single perceptron can not perfectly classify XOR because XOR is not linearly separable.

(3) Construct a network with two hidden neurons and one output neuron.

We can solve the XOR problem by using a small neural network.

Hidden neuron 1 works like an OR gate
Hidden neuron 2 works like and And gate
The output neuron combines their results to produce XOR.

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Input Layer
x_1, x_2

↓

Hidden neuron 1
$h_1 = H(x_1 + x_2 - 0.5)$
OR gate

Hidden Neuro 2
$h_2 = H(x_1 + x_2 - 1.5)$
AND gate

↓

Output Neuron
$y = H(h_1 - 2h_2 - 0.5)$
XOR Output

Hidden neuron-1: OR Gate.
formula.

$$h_1 = H(x_1 + x_2 - 0.5)$$

Weights and bias

$$w_1 = 1, w_2 = 1, b = -0.5$$

This neuron gives output 1 when at least one sensor is ON

Hidden neuron 2: AND gate.
formula.

$$h_2 = H(x_1 + x_2 - 1.5)$$

Weight and bias

$$w_1 = 1, w_2 = 1, b = -1.5$$

This neuron gives output 1 only when both sensors are ON.

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Output neuron
formula.

$$y = H(h_1 - 2h_2 - 0.5)$$

weights and bias.

$$w_{h_1} = 1, w_{h_2} = -2, b = -0.5$$

The negative weight -2 helps make the final output 0 when both hidden neurons are on.

Summary of all weights and biases

Neuron	Weights	Bias
Hidden neuron 1 (OR)	1, 1	-0.5
Hidden neuron 2 (AND)	1, 1	-1.5
Output neuron (XOR)	1, -2	-0.5

(4) Calculate the forward pass for all four samples.

A forward pass means calculating the output step-by-step. starting from the inputs and moving through the hidden neurons to the output neuron.

We use the same activation function for every neuron:

$$H(z) = \begin{cases} 1, & z \geq 0 \\ 0, & z < 0 \end{cases}$$

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Sample 1. $x_1 = 0, x_2 = 0$

Hidden neuron 1.

$$z_1 = 0 + 0 - 0.5 = -0.5$$
$$h_1 = H(-0.5) = 0$$

Hidden neuron 2.

$$z_2 = 0 + 0 - 1.5 = -1.5$$
$$h_2 = H(-1.5) = 0$$

Output neuron:

$$z_y = 0 - 2(0) - 0.5 = -0.5$$
$$y = H(-0.5) = 0$$

The predict output 0. correct.

Sample 2. $x_1 = 0, x_2 = 1$

Hidden neuron 1:

$$z_1 = 0 + 1 - 0.5 = +0.5$$
$$h_1 = H(0.5) = 1$$

Hidden neuron 2

$$z_2 = 0 + 1 - 1.5 = -0.5$$
$$h_2 = H(-0.5) = 0$$

Output neuron:

$$z_y = 1 - 2(0) - 0.5 = 0.5$$
$$y = H(0.5) = 1$$

The predicted output is 1 correct.

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Sample 3: $x_1 = 1, x_2 = 0$

Hidden neuron 1.

$$z_1 = 1 + 0 - 0.5 = 0.5$$
$$h_1 = H(0.5) = 1$$

Hidden neuron 2

$$z_2 = 1 + 0 - 1.5 = -0.5$$
$$h_2 = H(-0.5) = 0$$

Output neuron.

$$z_y = 1 - 2(0) - 0.5 = 0.5$$
$$y = H(0.5) = 1$$

The predicted output is 1 correct.

Sample 4: $x_1 = 1, x_2 = 1$.

Hidden neuron 1.

$$z_1 = 1 + 1 - 0.5 = 1.5$$
$$h_1 = H(1.5) = 1$$

Hidden neuron 2

$$z_2 = 1 + 1 - 1.5 = 0.5$$
$$h_2 = H(0.5) = 1$$

Output neuron.

$$z_y = 1 - 2(1) - 0.5 = -1.5$$
$$y = H(-1.5) = 0$$

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The predicted output 0 correct.

final truth truth.

x_1	x_2	h_1	h_2	Predicted y	Desired y
0	0	0	0	0	0
0	1	1	0	1	1
1	0	1	0	1	1
1	1	1	1	0	0

All four predictions are correct

$$\text{Accuracy} = \frac{4}{4} \times 100 = 100\ \%$$

(5) Why can increasing training from 100 to 10000 epochs fix the orginal Architecture?

first, understand two important terms.

A. Representablity.

Representablity means whether a model's architecture has the ability to present the required Solution

A single Perceptron creates a straight line disicion boundary

$$w_1x_1 + w_2x_2 + b = 0$$

But no single straight line can saparate the XOR out parts correctly.

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Therefore, a single perceptron can not represent the XOR function perfectly

B. Trainability

Trainablity means how easily a training algorithm can find suitable can find suitable weights weights and biases for a model.
However the orginal single perceptron has a different problem. no perfect solution exists within its architecture.

Number of epochs	Result
100	Can not Perfectly classify XOR
1000	Can not perfectly classify XOR
10,000	Can not Perfectly classify XOR
100 000	Can not Perfectly classify XOR

Increasing epochs gives the model more training iterations. But it does not change the Architecture.